EXP 27: Prolog Program to Implement Best First Search Algorithm

Name: Y. Roshitha (192424400)

Aim:

To develop a Prolog program that implements the Best First Search algorithm, so that at every step the neighbour with the lowest heuristic value is chosen to reach the goal.

Algorithm:

Step 1: Start the program.
Step 2: Represent the graph using nodes and edges.
Step 3: Assign heuristic values to all nodes.
Step 4: Select the starting node and goal node.
Step 5: Generate all neighboring nodes of the current node.
Step 6: Compare their heuristic values.
Step 7: Choose the node with the lowest heuristic value.
Step 8: Repeat the process until the goal node is reached.
Step 9: Display the path and stop.

Source Code:

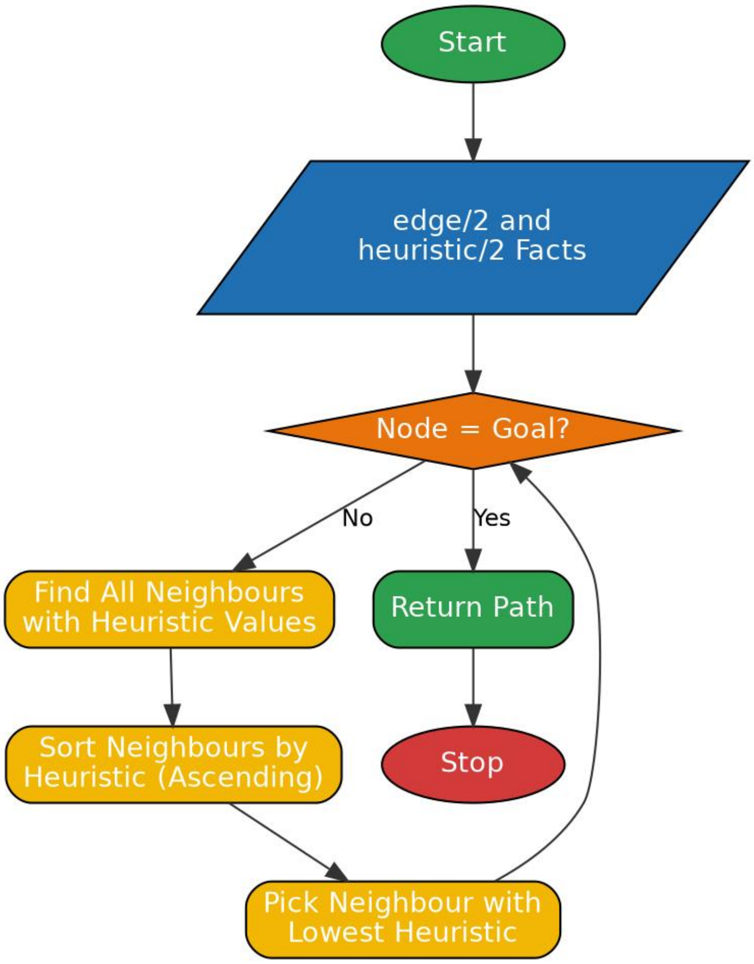
```prolog
% Program to implement Best First Search Algorithm

edge(a, b).
edge(a, c).
edge(b, d).
edge(b, e).
edge(c, f).
edge(e, g).
edge(f, g).

heuristic(a, 10).
heuristic(b, 8).
heuristic(c, 5).
heuristic(d, 7).
heuristic(e, 3).
heuristic(f, 4).
heuristic(g, 0).

best_first(Goal, Goal, [Goal]) :- !.
best_first(Node, Goal, [Node | Path]) :-
    findall(H-Next, (edge(Node, Next), heuristic(Next, H)), Pairs),
    keysort(Pairs, Sorted),
    member(_-Next, Sorted),
    best_first(Next, Goal, Path).
```

Flowchart:



Output:

```
Welcome to SWI-Prolog (threaded, 64 bits, version 9.2.9)
SWI-Prolog comes with ABSOLUTELY NO WARRANTY. This is free software.
Please run ?- license. for legal details.

For online help and background, visit https://www.swi-prolog.org
For built-in help, use ?- help(Topic). or ?- apropos(Word).

?- cd('C:/Users/yanar/Documents/CSA1717').
true.

?- [bestfirst].
true.

?- best_first(a, g, Path).
Path = [a, c, f, g] .
```

Result:

Thus the Prolog program to implement the Best First Search algorithm was written, executed successfully and the output was verified.